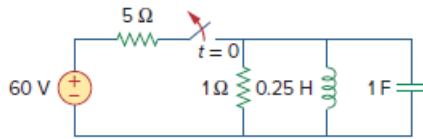


Problem

Find the voltage across the capacitor as a function of time for $t > 0$ for the circuit in Fig. 16.45. Assume steady-state conditions exist at $t = 0^-$.

Figure 16.45:



Step-by-step solution

Step 1 of 5

From the circuit,

For $t = 0$ sec, the switch is closed, the initial conditions for the voltage across capacitor and current through inductor are:

$$V(0) = 0 \text{ Volts}$$

and

$$i_L(0) = \frac{60}{5} \text{ Amperes}$$

$$i_L(0) = 12$$

Since the inductor acts as short circuit and capacitor acts as open circuit under steady state condition.

Apply the KCL to the source-free parallel RLC circuit

$$C \frac{dv(0)}{dt} + \frac{V(0)}{R} + I(0) = 0$$

$$\frac{dv(0)}{dt} = - \left(\frac{V(0) + RI(0)}{RC} \right)$$

$$\frac{dv(0)}{dt} = - \left(\frac{0 + 1 \cdot 12}{1 \cdot 1} \right)$$

$$\frac{dv(0)}{dt} = -12$$

Applying KCL to the source-free parallel RLC circuit for $t = 0$ sec

$$\frac{v(t)}{R} + \frac{1}{L} \int_{-\infty}^t v(t) dt + C \frac{dv(t)}{dt} = 0 \dots\dots (1)$$

Differentiate equation (1) with respect to 't'

$$\frac{1}{R} \frac{dv(t)}{dt} + \frac{v(t)}{L} + C \frac{d^2v(t)}{dt^2} = 0 \dots\dots (2)$$

Divide equation (2) by capacitance 'C'

$$\frac{d^2v(t)}{dt^2} + \frac{1}{RC} \frac{dv(t)}{dt} + \frac{v(t)}{LC} = 0 \dots\dots (3)$$

Step 2 of 5

We can now solve equation (3) to get the roots for the second order differential voltage equation that the solution is of exponential form

So, let $v(t) = Ae^{st}$ (4)

Where 'A' and 's' are constants to be determined

Substitute equation (4) into equation (3)

$$As^2e^{st} + \frac{A}{RC}e^{st} + \frac{A}{LC}e^{st} = 0$$

$$Ae^{st} \left(s^2 + \frac{1}{RC}s + \frac{1}{LC} \right) = 0$$

Implies that: $\left(s^2 + \frac{1}{RC}s + \frac{1}{LC} \right) = 0$ (5)

The equation (5) is a quadratic equation of the differential equation (3). Since the roots of the equation (5) dictate the character of $v(t)$

The two roots of an equation (5) are:

$$s_1 = \frac{-1}{2RC} + \sqrt{\left(\frac{1}{2RC} \right)^2 - \frac{1}{LC}} \text{ (6)}$$

$$s_2 = \frac{-1}{2RC} - \sqrt{\left(\frac{1}{2RC} \right)^2 - \frac{1}{LC}} \text{ (7)}$$

Equations (6) and (7) can be expressed in another way as:

$$s_1 = -\alpha + \sqrt{(\alpha)^2 - \omega_0^2} \text{ (8)}$$

$$s_2 = -\alpha - \sqrt{(\alpha)^2 - \omega_0^2}$$

The roots s_1 and s_2 are natural frequencies measured in nepers per second. Where ' α ' is a damping factor, ' ω_0 ' is a resonant frequency.

Step 3 of 5

When the switch is off (or open), we have a source-free RLC series circuit.

The expression for α is,

$$\alpha = \frac{1}{2RC}$$

Substitute 1 for R and 1 for C .

$$\alpha = \frac{1}{2 \times 1 \times 1}$$

$$\alpha = \frac{1}{2}$$

$$\alpha = 0.5$$

The expression for the resonant frequency is,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute 0.25 for L and 1 for C .

$$\omega_0 = \frac{1}{\sqrt{0.25 \times 1}}$$

$$\omega_0 = \frac{1}{0.5}$$

$$\omega_0 = 2$$

Since $\alpha < \omega_0$, we have the under-damped case

Step 4 of 5

The natural frequencies from equation (8) are: $s_{1,2} = -\alpha \pm \sqrt{(\alpha)^2 - \omega_0^2}$

$$s_{1,2} = -0.5 \pm \sqrt{0.5^2 - 2^2}$$

$$s_{1,2} = -0.5 \pm \sqrt{0.25 - 4}$$

$$s_{1,2} = -0.5 \pm \sqrt{-3.75}$$

$$s_{1,2} = -0.5 \pm j1.936$$

The equation (5) can be re-written as: $s^2 + 2\alpha s + \omega_0^2 = 0 \dots\dots (9)$

In this under-damped case the roots are complex and may be expressed as

$$s_{1,2} = -\alpha \pm j\omega_d$$

Where

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$\omega_d = \sqrt{2^2 - 0.5^2}$$

$$\omega_d = \sqrt{3.75}$$

$$\omega_d = 1.936$$

Thus, the natural response of source-free parallel RLC circuit is:

$$v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

$$v(t) = e^{-0.5t} (A_1 \cos 1.936t + A_2 \sin 1.936t) \dots\dots (10)$$

Apply the initial conditions to the equation (10)

$$v(0) = e^{-0.5 \times 0} (A_1 \cos 1.936 \times 0 + A_2 \sin 1.936 \times 0)$$

$$0 = 1(A_1 \cos 0 + A_2 \sin 0)$$

$$(A_1 \times 1 + A_2 \times 0) = 0$$

$$A_1 = 0$$

Here the constant $A_1 = 0$ the equation (10) can be re-written as:

$$v(t) = e^{-0.5t} (A_2 \sin 1.936t) \dots \dots (11)$$

Differentiate equation (11) with respect to 't'

$$\frac{dv(t)}{dt} = A_2 \times 1.936 \cos(1.936t) \times e^{-0.5t} + A_2 \times \sin(1.936t) \times -0.5e^{-0.5t} \dots \dots (12)$$

Apply the initial conditions to the equation (12)

$$\frac{dv(0)}{dt} = A_2 \times 1.936 \cos(1.936t) \times e^{-0.5t} + A_2 \times \sin(1.936t) \times -0.5e^{-0.5t}$$

$$-4 = A_2 \times 1.936 \cos(1.936 \times 0) \times e^{-0.5 \times 0} + A_2 \times \sin(1.936 \times 0) \times -0.5e^{-0.5 \times 0}$$

$$1.936A_2 = -4$$

$$A_2 = \frac{-4}{1.936}$$

$$A_2 = -2.066$$

Therefore, the voltage drop across the capacitor for $t > 0$ sec is,

$$v(t) = \boxed{-2.066e^{-0.5t} (\sin 1.936t)} \text{ Volts}$$

[Comments \(7\)](#)



Anonymous

lol



Anonymous

wtf is this



Anonymous

whos mans is this



Anonymous

LMAO



Anonymous

you the mans bro



Anonymous

fuck this, going to apply at Pizza hut bois, im outtie <3



Anonymous

This solution should have employed the laplace circuit method as it was a question from 16.22. Thanks for the solution though.